

Overall:

|                                                                      |    |
|----------------------------------------------------------------------|----|
| Did not follow submission instructions (per occurrence) <sup>1</sup> | -5 |
|----------------------------------------------------------------------|----|

**Q.1. Top-Down Design (40 points)<sup>2</sup>****Part 1 - Identifying Entities: (15 points)**

| <b>Error /Problem</b>                    | <b>Points lost</b> |
|------------------------------------------|--------------------|
| Missing 1 - 3 potential entities.        | -2 for each entity |
| Missing 4 or more entities.              | -8                 |
| Listed attributes as potential entities. | -2 for each error  |

**Part 2 - Identifying Attributes: (15 points)**

| <b>Error /Problem</b>                                                                     | <b>Points lost</b>    |
|-------------------------------------------------------------------------------------------|-----------------------|
| Missing an attribute.                                                                     | -1 for each attribute |
| Listed an attribute with the wrong entity.                                                | -1 for each attribute |
| Missing or incorrect creation of atomic parts for a multi-part data element, when needed. | -2 for each error     |

**Part 3 - Identifying Relationships: (10 points)**

| <b>Error /Problem</b>                                                                                                        | <b>Points lost</b>        |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 1 - 3 potential relationships are missing.                                                                                   | -2 for each relationship  |
| Potential relationship is incorrect (i.e., a 1-to-many relationship is listed as a many-to-many relationship or visa versa). | -2 for each relationship  |
| 4 or more potential relationships are missing or incorrect.                                                                  | -8                        |
| Primary (PK) or foreign (FK) key attributes are missing or incorrect                                                         | -1 for each missing key   |
| A PK/FK attribute is not labeled as such.                                                                                    | -1 for each missing label |

Overall:

| <b>Error /Problem</b>                             | <b>Points lost</b>      |
|---------------------------------------------------|-------------------------|
| Question unanswered.                              | -40                     |
| Other outstanding error. (Depending on severity). | -1 to -3 for each error |

***See next page for Question 2 rubric***

<sup>1</sup> Including, but not limited to, failure to follow file type and upload instructions. These points will be deducted from the overall grade for this assignment after individual questions are graded and will be applied in addition to grade reductions on individual questions.

<sup>2</sup> Points lost might add up to more than available points, in which case the answer is graded as zero.

## Q.2. Normalization (60 points)<sup>3</sup>

### 1NF (20 points):

| Error /Problem                                                                                                                                                                 | Points lost               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Repeating group was not removed from the original table.                                                                                                                       | -3 for each group         |
| The repeating fields are not collapsed into single fields. (There are repeating fields in the newly-created table.)                                                            | -3 for each group         |
| A necessary PK field is not added to the newly-created table.                                                                                                                  | -3 for each PK field      |
| A PK/CPK/FK field is not labeled as such.                                                                                                                                      | -1 for each missing label |
| A relationship line/arrow is not drawn between the PK of the original table, and the CPK or FK field in the newly-created table corresponding to the PK of the original table. | -1 for each missing line  |
| 2NF tasks are included in 1NF.                                                                                                                                                 | -6                        |
| 3NF tasks are included in 1NF.                                                                                                                                                 | -6                        |
| Missing/incorrect description for 1NF.                                                                                                                                         | -1 to -5                  |

### 2NF (20 points):

| Error /Problem                                                                                                         | Points lost               |
|------------------------------------------------------------------------------------------------------------------------|---------------------------|
| One of the fields that pose a partial dependency is not moved to a separate table.                                     | -3 for each field         |
| One of the fields that do not pose a partial dependency is incorrectly moved to a separate table.                      | -3 for each field         |
| A table that potentially includes partial dependencies (a table with a CPK structure) is not included in 2NF analysis. | -6                        |
| A PK/CPK/FK field is not labeled as such.                                                                              | -1 for each missing label |
| A relationship line/arrow is not drawn between a PK/FK (or CPK).                                                       | -1 for each missing line  |
| 1NF tasks are included in 2NF.                                                                                         | -6                        |
| 3NF tasks are included in 2NF.                                                                                         | -6                        |
| Missing/incorrect description for 2NF.                                                                                 | -1 to -5                  |

### 3NF (20 points):

| Error /Problem                                                                                       | Points lost               |
|------------------------------------------------------------------------------------------------------|---------------------------|
| One of the fields that pose a transitive dependency is not moved to a separate table.                | -3 for each field         |
| One of the fields that do not pose a transitive dependency is incorrectly moved to a separate table. | -3 for each field         |
| A table that potentially includes transitive dependencies is not included in 3NF analysis.           | -6                        |
| A PK/CPK/FK field is not labeled as such.                                                            | -1 for each missing label |
| A relationship line/arrow is not drawn between a PK/FK (or CPK).                                     | -1 for each missing line  |
| 1NF tasks are included in 3NF.                                                                       | -6                        |
| 2NF tasks are included in 3NF.                                                                       | -6                        |
| Missing/incorrect description for 3NF.                                                               | -1 to -5                  |

### Overall:

| Error /Problem                                    | Points lost             |
|---------------------------------------------------|-------------------------|
| Question unanswered.                              | -60                     |
| Other outstanding error. (Depending on severity). | -2 to -8 for each error |

<sup>3</sup> Points lost might add up to more than available points, in which case the answer is graded as zero.